

Extra Hidden States under Emission Misspecification in Hidden Markov Models

Aurélien Nicosia

Département de mathématiques et de statistique, Université Laval

Québec, Québec, Canada

`aurelien.nicosia@mat.ulaval.ca`

ORCID: [0000-0002-0974-9988](https://orcid.org/0000-0002-0974-9988)

Abstract

Additional states in a hidden Markov model can represent scientific regimes, but they can also compensate for restrictive state-dependent distributions. We formalize the latter mechanism. When aggregate-state emissions are identifiable finite mixtures, a refined model with simple emissions reproduces the observed process exactly, whereas a model with only the aggregate number of simple-emission states remains separated from the truth in relative-entropy rate. This result motivates an applied diagnostic workflow that compares a simple-emission model, the same model with an extra state, and a model with the original state count but richer emissions. One-step predictive calibration and decoded-state mappings complement information criteria. Simulations recover the predicted state-splitting mechanism. In daily movement data from four Canadian elk, a third Gaussian state improves on two Gaussian states, but a two-state Gaussian-mixture model is competitive for all animals and preferred by BIC for three. The findings show how practitioners can distinguish useful emission corrections from evidence for additional scientific regimes.

Keywords: hidden Markov model; emission misspecification; finite mixture; relative-entropy rate; state splitting; movement ecology.

MSC 2020: Primary 62M05; Secondary 62M10.

1 Introduction

Hidden Markov models (HMMs) separate serial dependence from the conditional distribution of observations through a latent state process. This decomposition supports applications across time-series analysis, signal processing, and ecology [Rabiner, 1989, Cappé et al., 2005, Zucchini et al., 2016]. Fitted states are often labelled as behaviours, regimes, phases, or classes and then become the main scientific object reported from an analysis [McClintock et al., 2020].

That interpretation is useful but not automatic. A fitted state is also part of a statistical approximation. If the state-dependent distribution is too restrictive, likelihood can improve by

creating states whose emissions jointly approximate skewness, heavy tails, overdispersion, or within-regime heterogeneity. An additional state can therefore be statistically useful without representing an additional scientific regime. Under misspecification, likelihood targets a best approximation in the fitted family rather than the data-generating parameter [White, 1982, Douc and Moulines, 2012].

The practical phenomenon is established in the HMM literature. Pohle et al. [2017] show that information criteria can favour state counts that are difficult to interpret scientifically. Additional states may absorb unmodelled structure in state-dependent distributions, while flexible parametric, mixture, and nonparametric observation models offer alternatives [Dannemann, 2012, Dannemann et al., 2014, Langrock et al., 2015, 2018]. The distinction matters because methods for estimating statistical HMM order [Gassiat and Keribin, 2000, Gassiat and Rousseau, 2014, Lehericy, 2019, Zou et al., 2024, Lin and Huang, 2025] do not imply that every selected state is a distinct scientific regime.

This article makes the mechanism precise and translates it into a practical workflow. At the theoretical level, a state-space refinement reproduces a persistent aggregate HMM with mixture emissions exactly. Predictive mixture geometry then shows that every model with only the aggregate number of simple emissions remains separated from the true observed process by a positive relative-entropy rate. The comparison is made between observed process laws, so no alignment of candidate and true hidden trajectories is assumed. The result complements HMM likelihood and identifiability theory [Leroux, 1992, Bickel et al., 1998, Allman et al., 2009, Gassiat et al., 2016, Alexandrovich et al., 2016].

For practice, we propose comparing three fits: a scientifically plausible state count with simple emissions, an additional-state version of that model, and the original state count with enriched emissions. The Bayesian information criterion (BIC) is used together with one-step predictive calibration, decoded paths, and a direct mapping between the two latent partitions. A simulation isolates the mechanism. A real-data analysis of four Canadian elk then shows what the comparison reveals in a familiar movement setting. The aim is not to determine a latent ontology from data alone, but to give practitioners tangible evidence before an extra state receives a scientific label.

Section 2 gives the formal mechanism; Sections 3 and 4 describe its estimation and practical diagnosis; and Sections 5 and 6 illustrate the comparison in simulation and data.

2 Why restrictive emissions create extra states

The practical message of this section is simple. If a scientifically meaningful regime has a heterogeneous emission distribution, several simple-emission states can improve fit by representing its components rather than by revealing new regimes. The formal results below give conditions under which an aggregate-state model with richer emissions and a refined model with additional simple-emission states generate the same observed process.

Let $\mathbf{Y}$ be the observation set, let $\mathcal{B}$ be its sigma-field, let λ be a dominating measure on $(\mathbf{Y}, \mathcal{B})$,

and let Θ be the emission-parameter space. All densities below are with respect to λ . For densities p and q , define the Kullback–Leibler divergence and the $L^1(\lambda)$ distance by

$$D(p\|q) = \int_Y p(y) \log \left\{ \frac{p(y)}{q(y)} \right\} \lambda(dy), \quad \|p - q\|_1 = \int_Y |p(y) - q(y)| \lambda(dy).$$

Here, $0 \log(0/q) = 0$, and $D(p\|q) = +\infty$ if $q = 0$ on a set having positive probability under p . The same notation $D(P\|Q)$ is used for probability laws through their densities. Let $\mathcal{F} = \{f_\eta : \eta \in \Theta\}$ be the simple fitted emission family. Define the class of mixtures with at most K atoms, for an integer $K \geq 1$, by

$$\mathcal{M}_K = \left\{ \sum_{j=1}^K \omega_j f_{\eta_j} : \omega_j \geq 0, \sum_{j=1}^K \omega_j = 1, \eta_j \in \Theta \right\}. \quad (1)$$

Thus, the ω_j are mixing proportions. Zero weights are permitted, so $\mathcal{M}_K \subseteq \mathcal{M}_{K+1}$. Let $\mathbf{H}_K(\mathcal{F})$ denote the observed process laws generated by K -state HMMs with emissions in $\mathcal{F}$, arbitrary transition matrices, and arbitrary initial distributions. For process laws P and Q , let $P^{1:T}$ and $Q^{1:T}$ denote the respective joint laws of $(Y_1, \dots, Y_T)$ and write

$$\underline{D}^{\text{rate}}(P\|Q) = \liminf_{T \rightarrow \infty} T^{-1} D(P^{1:T}\|Q^{1:T})$$

for their lower relative-entropy rate.

Fix an integer $K_0 \geq 1$, interpreted as the number of aggregate scientific regimes. A superscript 0 denotes a data-generating parameter. The following assumptions isolate a transparent finite-mixture setting.

Assumption 2.1 (Emission regularity). The parameter space Θ is compact, the map $\eta \mapsto f_\eta$ is continuous into $L^1(\lambda)$, and every f_η is positive λ -almost everywhere.

Assumption 2.2 (Identifiable mixture order). Finite mixtures from $\mathcal{F}$ have identifiable minimal order. Two equal finite mixtures with positive weights and pairwise distinct parameters have the same number of components and agree up to permutation.

Classical sufficient conditions are given by [Teicher \[1963\]](#) and [Yakowitz and Spragins \[1968\]](#); see also [McLachlan and Peel \[2000\]](#). The assumptions include univariate Gaussian emissions on compact mean and variance sets with variances bounded away from zero.

Lemma 2.3 (Separation from lower-order mixtures). *Under Assumptions 2.1 and 2.2, let $R > K_0$ be an integer and let $\eta_1^0, \dots, \eta_R^0$ be pairwise distinct. For $c > 0$ and $Rc \leq 1$, set*

$$\mathcal{A}_c = \left\{ \sum_{r=1}^R \omega_r f_{\eta_r^0} : \omega_r \geq c, \sum_{r=1}^R \omega_r = 1 \right\}.$$

Then $d_c := \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} \|p - q\|_1 > 0$, and

$$\inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} D(p\|q) \geq d_c^2/2 > 0. \quad (2)$$

The lemma says that a mixture containing R distinct atoms, each with a non-negligible weight, cannot be approximated arbitrarily closely by only $K_0 < R$ atoms.

To see the approximation incentive directly, let $Z_t \in \{1, \dots, K_0\}$ be a stationary aggregate Markov chain with transition matrix $\Gamma^0 = (\gamma_{k\ell}^0)$ and stationary probabilities $\pi_k = \mathbb{P}(Z_t = k) > 0$. Here, $\gamma_{k\ell}^0 = \mathbb{P}(Z_t = \ell \mid Z_{t-1} = k)$. Suppose the emission in state k is g_k , and, for an integer $m \geq 1$, define $\delta_k(m) = \inf_{q \in \mathcal{M}_m} D(g_k \| q)$. Thus, $\delta_k(m)$ is the smallest emission-level Kullback–Leibler loss attainable with at most m atoms from $\mathcal{F}$.

Proposition 2.4 (Aligned improvement by state splitting). *If aggregate state k is represented by one emission from $\mathcal{F}$, the best aligned Kullback–Leibler risk is $\sum_{k=1}^{K_0} \pi_k \delta_k(1)$. If it is represented by $m_k \geq 1$ substates, where m_k is the number allocated to aggregate state k , the best aligned risk is $\sum_{k=1}^{K_0} \pi_k \delta_k(m_k)$. The second risk is strictly smaller whenever $\delta_{k^*}(m_{k^*}) < \delta_{k^*}(1)$ for at least one state.*

In plain terms, splitting gives a heterogeneous state several emission atoms instead of one. The statement is only an aligned comparison because it holds the aggregate trajectory fixed. The next construction removes that restriction.

Suppose each aggregate emission is a finite mixture with m_k components,

$$g_k(y) = \sum_{a=1}^{m_k} \omega_{k,a} f_{\eta_{k,a}^0}(y), \quad \omega_{k,a} > 0, \quad \sum_a \omega_{k,a} = 1. \quad (3)$$

The $\omega_{k,a}$ are the within-state mixing proportions.

Proposition 2.5 (Exact HMM refinement). *Let the refined states be (k, a) , where $1 \leq k \leq K_0$ and $1 \leq a \leq m_k$. Give state (k, a) emission $f_{\eta_{k,a}^0}$ and transition probabilities*

$$\Gamma_{(k,a),(\ell,b)}^{\text{ref}} = \gamma_{k\ell}^0 \omega_{\ell,b}. \quad (4)$$

With initial probabilities $\pi_k \omega_{k,a}$, the refined HMM has exactly the same observed process law as the aggregate HMM with transitions Γ^0 and emissions in (3).

Equation (4) retains the probability of moving from aggregate state k to ℓ , then draws a component of the destination emission. The refined states therefore encode within-regime mixture components, not new aggregate dynamics.

Assume now that $\varepsilon := \min_{1 \leq k, \ell \leq K_0} \gamma_{k\ell}^0 > 0$, let $R = \sum_{k=1}^{K_0} m_k$, let $\omega_{\min} = \min_{1 \leq k \leq K_0, 1 \leq a \leq m_k} \omega_{k,a}$, and set $c = \varepsilon \omega_{\min}$. Suppose $R > K_0$ and all $\eta_{k,a}^0$ are pairwise distinct.

Theorem 2.6 (Observed-law separation). *Under Assumptions 2.1 and 2.2, the stationary aggregate HMM with emissions (3) has observed law P_0^Y satisfying*

$$\inf_{Q \in \mathcal{H}_R(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0, \quad \inf_{Q \in \mathcal{H}_{K_0}(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) \geq d_c^2/2 > 0. \quad (5)$$

The first equality follows because the refined R -state model reproduces the truth exactly. For the second, every one-step prediction from a K_0 -state simple-emission HMM belongs to $\mathcal{M}_{K_0}$, while every true prediction contains all R atoms with weights at least c . The mixture separation lemma and the chain rule for relative entropy then give the positive rate gap. This argument concerns observed predictive distributions and does not align hidden paths.

Corollary 2.7 (Enriched-emission representation). *If the fitted emission family is enlarged from $\mathcal{F}$ to $\mathcal{M}_M$, with an integer $M \geq \max_{1 \leq k \leq K_0} m_k$, then*

$$\inf_{Q \in \mathcal{H}_{K_0}(\mathcal{M}_M)} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0.$$

Thus, a refined model with more simple-emission states and an aggregate model with richer emissions can describe the same observations. Equality of the two representations does not determine which latent description is scientifically meaningful.

For reference, Sections S1.1–S1.5 of the Supporting Information give proofs of Lemma 2.3, Proposition 2.4, Proposition 2.5, Theorem 2.6, and Corollary 2.7, respectively.

3 Fitting the enriched-emission HMM

Let $X_t \in \{1, \dots, K\}$ denote the fitted hidden state at time t . For state k , let $M_k \geq 1$ be the number of fitted Gaussian components and let $\phi(y; \mu, \sigma^2)$ denote the Gaussian density at y with mean μ and variance $\sigma^2 > 0$. The fitted state-dependent density is

$$b_k(y) = \sum_{m=1}^{M_k} \omega_{km} \phi(y; \mu_{km}, \sigma_{km}^2), \quad \omega_{km} \geq 0, \quad \sum_m \omega_{km} = 1. \quad (6)$$

The quantities ω_{km} are mixing proportions within state k ; they do not change the state transition probabilities. The hidden chain has initial probabilities $\alpha_k = \mathbb{P}(X_1 = k)$ and transition probabilities $a_{k\ell} = \mathbb{P}(X_t = \ell \mid X_{t-1} = k)$, collected in $A = (a_{k\ell})$.

The expectation-maximization (EM) algorithm alternates between state and component allocation and closed-form parameter updates. Write $Y_{1:T} = (Y_1, \dots, Y_T)$ and let $C_t \in \{1, \dots, M_{X_t}\}$ denote the Gaussian-component label at time t . A scaled forward–backward recursion gives

$$\begin{aligned} \zeta_t(k) &= \mathbb{P}(X_t = k \mid Y_{1:T}), \\ \xi_t(k, \ell) &= \mathbb{P}(X_{t-1} = k, X_t = \ell \mid Y_{1:T}). \end{aligned}$$

Here, $\zeta_t(k)$ is defined for $t = 1, \dots, T$, whereas $\xi_t(k, \ell)$ is defined for $t = 2, \dots, T$. The joint posterior responsibility for state k and component m is

$$r_t(k, m) = \mathbb{P}(X_t = k, C_t = m \mid Y_{1:T}) = \zeta_t(k) \frac{\omega_{km} \phi(Y_t; \mu_{km}, \sigma_{km}^2)}{b_k(Y_t)}. \quad (7)$$

The M-step updates are

$$\alpha_k^{\text{new}} = \zeta_1(k), \quad a_{k\ell}^{\text{new}} = \frac{\sum_{t=2}^T \xi_t(k, \ell)}{\sum_{t=2}^T \sum_j \xi_t(k, j)}, \quad (8)$$

$$\omega_{km}^{\text{new}} = \frac{\sum_t r_t(k, m)}{\sum_t \zeta_t(k)}, \quad \mu_{km}^{\text{new}} = \frac{\sum_t r_t(k, m) Y_t}{\sum_t r_t(k, m)}, \quad (9)$$

$$(\sigma_{km}^2)^{\text{new}} = \frac{\sum_t r_t(k, m) (Y_t - \mu_{km}^{\text{new}})^2}{\sum_t r_t(k, m)}. \quad (10)$$

In these updates, sums over t without displayed limits run from 1 to T , and $\sum_j$ runs over the K fitted states. In computation, a small lower bound on σ_{km} prevents likelihood degeneracy. Multiple starts are needed because mixture-HMM likelihoods are multimodal. For a sequence of length T , BIC is $-2\ell(\hat{\theta}) + p \log T$, where $\hat{\theta}$ is the maximum likelihood estimate, $\ell(\hat{\theta})$ is the maximized log-likelihood, and the number p of free parameters is

$$(K - 1) + K(K - 1) + \sum_{k=1}^K \{(M_k - 1) + 2M_k\}. \quad (11)$$

The three terms count initial probabilities, transition probabilities, and state-specific mixture parameters, respectively.

4 A diagnostic workflow for practitioners

Suppose scientific knowledge suggests K_0 regimes, while a simple-emission analysis favours $K_0 + 1$ states. We recommend the following comparison.

1. Fit the simple-emission HMMs with K_0 and $K_0 + 1$ states, then fit a K_0 -state model with a scientifically plausible richer emission family.
2. Compare penalized fit, but do not use BIC alone as evidence for a new scientific regime.
3. Check one-step predictive calibration. For continuous data, compute the probability integral transform (PIT) values $U_t = F_t(Y_t \mid Y_{1:t-1})$, where F_t is the fitted one-step predictive cumulative distribution function. A calibrated model should produce PIT values approximately uniform on $(0, 1)$.
4. Compare decoded sequences and state-dependent distributions. Cross-tabulate the $K_0 + 1$ -state decoding against the enriched K_0 -state decoding. If two simple-emission states map largely to one enriched state, they are candidates for an emission-driven split.
5. Present the observed series, predictive diagnostics, and mapping to the context researcher. Interpret an extra state substantively only when its temporal and domain-specific meaning remains after plausible emission enrichment.

These steps do not prove that an extra state is corrective. They provide direct, data-related evidence that is easier to discuss than a BIC difference alone. Formal goodness-of-fit procedures, sensitivity to starting values, and subject-matter validation remain important.

5 Simulation study

We first isolate the mechanism in a controlled setting. The data-generating process has two aggregate states, transition matrix

$$\Gamma^0 = \begin{pmatrix} 0.93 & 0.07 \\ 0.05 & 0.95 \end{pmatrix}.$$

The first-state emission was a two-component Gaussian mixture with weights 0.70 and 0.30, means -1.20 and 1.10 , and standard deviations 0.55 and 0.45 . The second-state emission was Gaussian with mean 3.80 and standard deviation 0.65 . The process therefore has two aggregate regimes and three distinct Gaussian atoms.

For a reference series of length 1500, we fitted Gaussian HMMs with $K = 2$ and $K = 3$, and an enriched $K = 2$ HMM with two Gaussian components per state. Each model used 10 EM starts. The two-state Gaussian model merged the heterogeneous first regime and had BIC 4487.4. The three-state Gaussian model separated its two components and had BIC 4197.3. The enriched two-state model recovered the same within-regime structure while retaining two aggregate states and had BIC 4191.8. [Figure 1](#) shows the directly observable emission fit.

Across 200 fixed-seed replications at each sample size, BIC selected the third Gaussian state in 89.5%, 94.5%, and 96.0% of samples for $T = 500, 1500, 5000$, respectively. When the three-state Gaussian model was compared with the enriched two-state model, BIC preferred the latter in 94.0%, 98.5%, and 98.5% of samples. Full computational details and additional diagnostics are reported in the Supporting Information.

These frequencies are descriptive. Finite mixtures and overfitted HMMs contain singular parameter strata, so regular-model interpretations of BIC require caution [[Keribin, 2000](#), [Rousseau and Mengersen, 2011](#), [Gassiat and Rousseau, 2014](#), [Watanabe, 2013](#), [Drton and Plummer, 2017](#)].

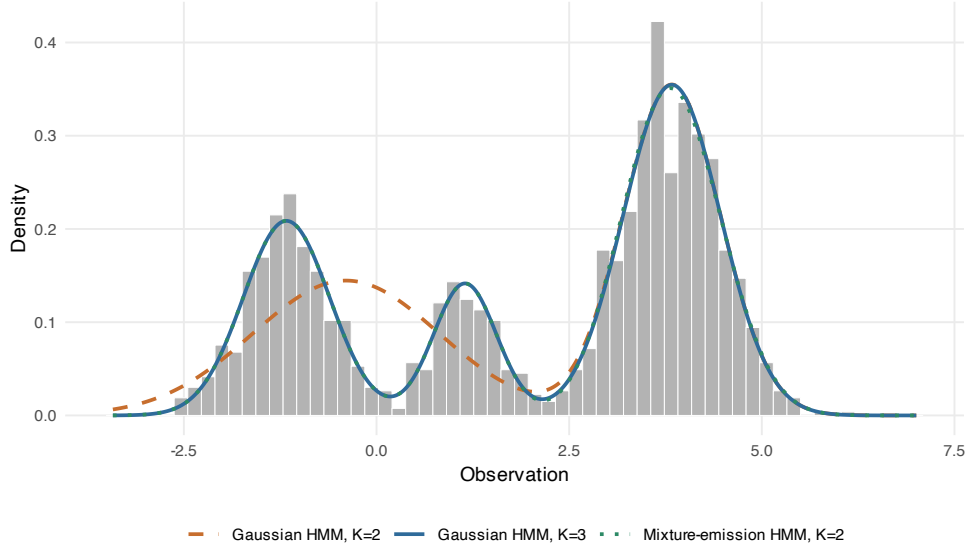


Figure 1: Observed histogram and fitted marginal densities for the reference simulation. The two-state Gaussian model smooths over the heterogeneous first regime. The three-state Gaussian and enriched two-state models recover its two components.

6 Canadian elk movement application

6.1 Data and analysis

We used the public `elk_data` distributed with the R package `moveHMM` [Michelot et al., 2016]. The data contain 735 ordered GPS locations for four elk in east-central Ontario, with 194, 159, 164, and 218 locations for elk 115, 163, 287, and 363, respectively. The animals and study setting originate from Morales et al. [2004]. Exact timestamps are absent from the distributed data; following the documented use of elk 115 in Parton and Blackwell [2017], consecutive locations were treated as approximately 24 hours apart. We calculated 193, 158, 163, and 217 daily step lengths.

The analysis deliberately uses only step length. It is not a full ecological reanalysis and does not assign behavioural labels. Its purpose is to show how emission flexibility changes the interpretation of fitted state count. For each animal separately, the response was $\log(1 + \text{step length in km})$. We fitted Gaussian HMMs with $K = 2$ and $K = 3$, plus an enriched $K = 2$ HMM with two Gaussian components in each state-dependent distribution. Separate fits avoid imposing identical movement distributions across animals. Each fit used 200 deterministic-seed starts, a fitted standard-deviation lower bound of 0.05, and the same screening and convergence rules. The models have 7, 14, and 13 free parameters, respectively.

We compared BIC and the empirical distribution of in-sample one-step PIT values. Because the same observations are used to estimate and assess each model, these PIT summaries are descriptive calibration diagnostics rather than out-of-sample performance estimates. We also compared Viterbi-decoded state sequences, meaning the most likely joint hidden-state paths, and cross-tabulated the

Table 1: Empirical comparison for four Canadian elk. Lower BIC is better. The enriched model has two aggregate states and two Gaussian components in each state-dependent distribution.

Animal	steps	Gaussian $K = 2$	Gaussian $K = 3$	Mixture $K = 2$	ΔBIC
elk-115	193	245.8	230.3	233.1	+2.8
elk-163	158	128.4	108.9	98.1	-10.8
elk-287	163	149.0	83.9	75.2	-8.7
elk-363	217	197.6	165.8	157.9	-8.0

Note: ΔBIC is mixture $K = 2$ minus Gaussian $K = 3$; negative values favour the two-state enriched-emission model.

three Gaussian states against the two enriched-emission states.

6.2 Results

The three-state Gaussian model improved substantially on the two-state Gaussian model for every animal, with BIC reductions from 15.5 to 65.1 [Table 1](#). That result alone could be presented as evidence for a third movement state. The enriched comparison changes the interpretation. The two-state mixture model had lower BIC than the three-state Gaussian model for elk 163, 287, and 363, by 10.8, 8.7, and 8.0 points. For elk 115 its BIC was 2.8 points higher, so the two representations were close on that criterion.

The enriched two-state model also had the smallest PIT distance for elk 115, 287, and 363, and was close to the three-state Gaussian model for elk 163 [Figure 2](#). Its central 90% predictive interval coverage was between 0.914 and 0.931 across animals. The corresponding range was 0.914 to 0.948 for the three-state Gaussian model. Thus, reducing the number of states after enriching the emissions did not produce a general loss of predictive calibration.

The fitted component distributions show what the enrichment contributes beyond a third Gaussian state. For elk 115, one enriched state combines component medians of 2.60 and 9.39 km with weights 0.70 and 0.30. The corresponding pairs are 0.58 and 5.26 km for elk 163, 0.11 and 3.77 km for elk 287, and 0.13 and 4.43 km for elk 363. Thus, a single aggregate state can accommodate distinct parts of the observed step-length distribution without assigning each part a separate state. These components are distributional features, not behavioural categories; complete component and transition estimates are reported in the Supporting Information.

The only near-boundary transition estimates occurred for elk 287: the two self-transition probabilities were 1.000 and 0.989. Inference about aggregate switching for that individual should therefore be treated with particular caution.

The decoded-state mapping makes the mechanism tangible [Figure 3](#). For elk 115, the first two Gaussian states mapped almost entirely to the first enriched state, while the third mapped to the second. For elk 287, the first and third Gaussian states mapped to one enriched state and the second mapped to the other, up to label permutation. Elk 163 showed the same broad aggregation with partial overlap in one Gaussian state. The mapping for elk 363 was less discrete, indicating that the two latent descriptions should not be treated as interchangeable for that animal.

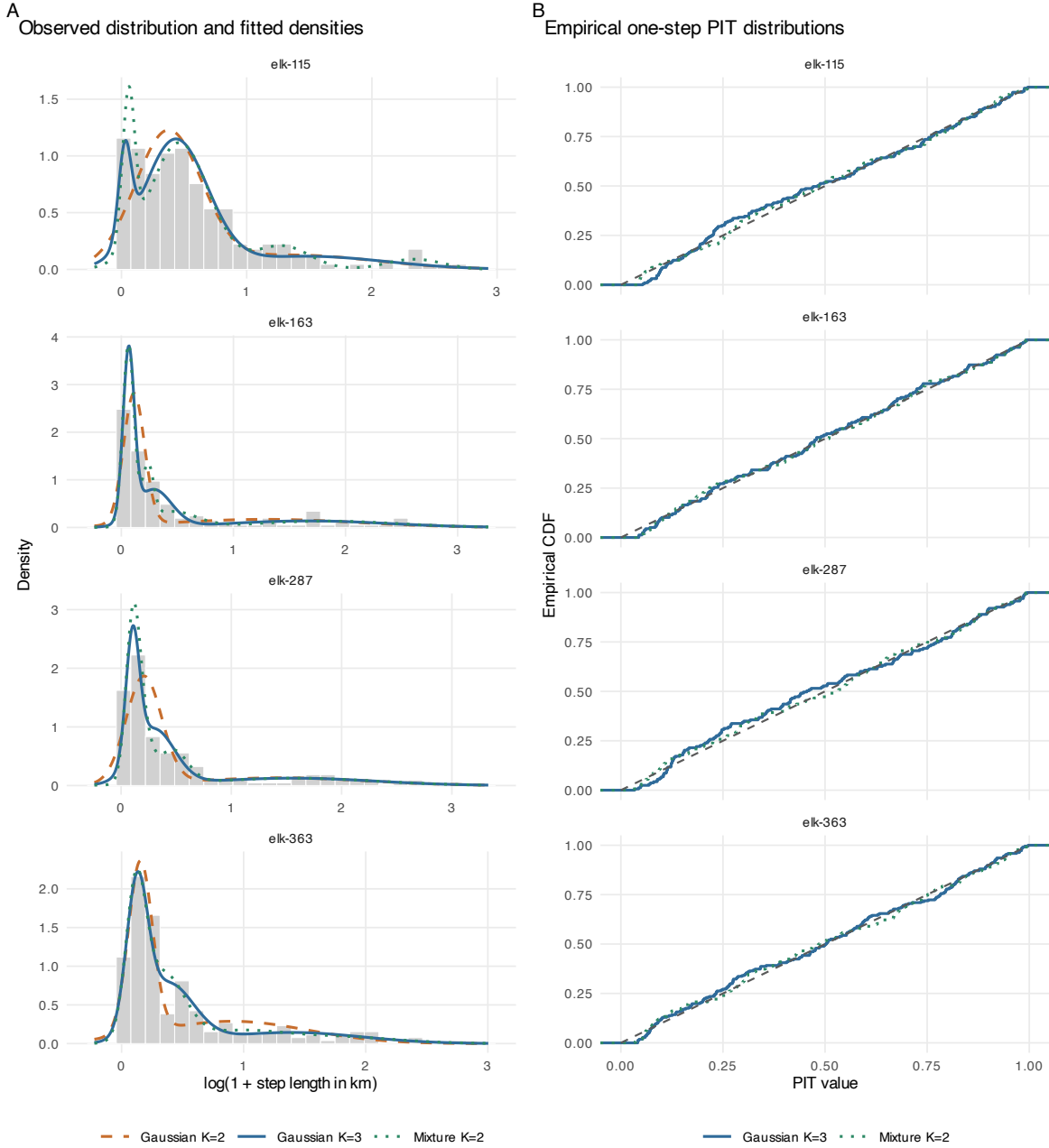


Figure 2: Diagnostics directly connected to the observed elk step lengths. Panel A overlays the histogram of the transformed observations with fitted emission densities weighted by posterior state occupation. Panel B compares the empirical one-step PIT distributions of the Gaussian $K = 3$ and enriched $K = 2$ models with the uniform reference. Agreement with the histogram and the diagonal are descriptive checks of distributional fit and predictive calibration, respectively.

Mapping of three Gaussian states to two enriched-emission states

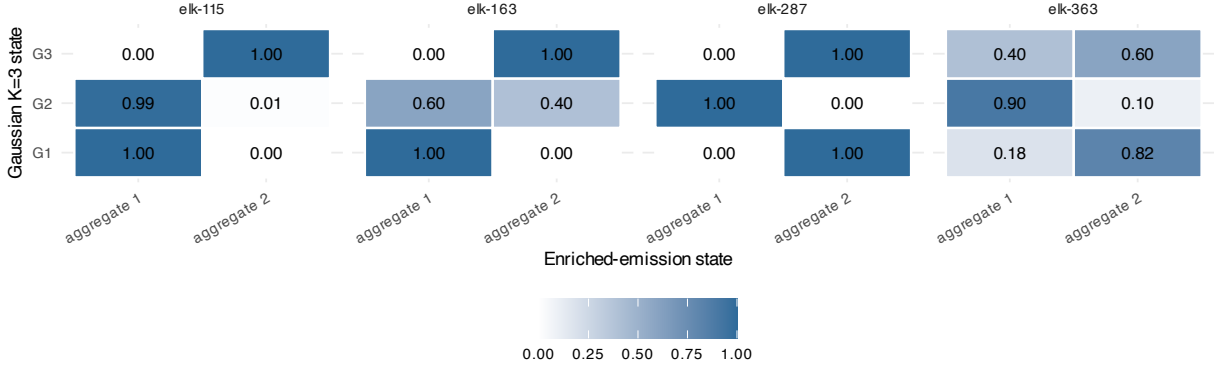


Figure 3: Row proportions from cross-tabulations of Viterbi states. Each panel shows how observations decoded into each state of the Gaussian $K = 3$ model are allocated between the two states of the enriched-emission model. State labels are model-specific and have no behavioural interpretation here.

Taken together, the observation-level diagnostics support an emission-driven explanation for at least part of the third Gaussian state. They do not establish that elk movement has exactly two biological behaviours. Instead, they show that the evidence for a third state is sensitive to the state-dependent distribution and should not be presented to a context researcher as a new behaviour on BIC alone.

7 Discussion

The theoretical and empirical results clarify why additional HMM states can be both useful and scientifically misleading. They are useful when they improve the approximation of a restrictive observation model. They are misleading when approximation components are labelled as new regimes without testing plausible emission alternatives.

The proposed workflow makes that test directly visible. In the elk example, three Gaussian states fit better than two, but two richer state-dependent distributions match or improve the penalized fit for three animals and improve the PIT distance for three. Decoded-state cross-tabulations show where the third Gaussian partition collapses after enrichment. These displays are more tangible than a BIC table alone, yet they remain diagnostics rather than an automatic state-merging rule.

Several limitations are important. Extra states can correct emissions, transitions, missing covariates, temporal resolution, or multiple mechanisms at once. Equality or near-equality of observed-law representations cannot identify a unique latent ontology. Domain knowledge, robustness to covariates and dependence structures, and scientific validation are therefore necessary [Pohle et al., 2017, McClintock et al., 2020]. The theorem is a one-way warning: under its assumptions, the aggregate-state simple-emission model is necessarily misspecified.

The exact result assumes finite mixtures and a positive aggregate transition matrix. Heavy-tailed

or skewed emissions may require successively richer approximations rather than a finite exact refinement. Structural zeros in the transition matrix would require blockwise accessibility conditions instead of the uniform one-step bound used here.

The elk illustration has additional limitations. The public data lack exact timestamps, the sample sizes per animal are modest, only step length is used, and the PIT checks are in-sample. Direction, habitat covariates, individual heterogeneity, and irregular observation times are outside its scope. The analysis consequently demonstrates a diagnostic workflow, not a definitive behavioural classification.

8 Conclusion

Emission misspecification can make additional hidden states valuable for observed-law approximation even when the scientific process has fewer aggregate regimes. Exact refinement and predictive mixture separation explain the mechanism. Comparing simple additional-state models with enriched-emission models, then examining predictive calibration and decoded-state mappings, gives practitioners concrete evidence before assigning substantive meaning to an extra state.

Acknowledgments

During preparation of the manuscript and supporting computational materials, the author used OpenAI ChatGPT and Codex for language editing, manuscript restructuring, code development and organization, and reproducibility checks. These tools were not treated as authors or sources of evidence. The author reviewed and edited all tool-assisted material; regenerated the reported numerical results from the archived code; and takes full responsibility for the content, accuracy, and integrity of the article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability statement

The elk locations are distributed as `elk_data` in the open-source R package `moveHMM`. The reproducibility package accompanying this submission, named `4416781_Reproducibility_Package.zip`, contains the package version, source references, in-memory object digest, analysis code, fixed seeds, and generated outputs, including row-level simulation results. The workflow is implemented entirely in R, reads `elk_data` directly from the package, and does not redistribute the raw locations. The same code and derived outputs are publicly available at <https://github.com/AurelienNicosia/ULaval/corrective-hmm-states>.

Ethics statement

This study is a secondary analysis of publicly distributed data. No new data were collected and no animals were handled for this study.

Conflict of interest statement

The author declares no conflict of interest.

References

- Grigory Alexandrovich, Hajo Holzmann, and Anna Leister. Nonparametric identification and maximum likelihood estimation for hidden Markov models. *Biometrika*, 103(2):423–434, 2016. doi: 10.1093/biomet/asw001.
- Elizabeth S. Allman, Catherine Matias, and John A. Rhodes. Identifiability of parameters in latent structure models with many observed variables. *The Annals of Statistics*, 37(6A):3099–3132, 2009. doi: 10.1214/09-AOS689.
- Peter J. Bickel, Ya’acov Ritov, and Tobias Rydén. Asymptotic normality of the maximum-likelihood estimator for general hidden Markov models. *The Annals of Statistics*, 26(4):1614–1635, 1998. doi: 10.1214/aos/1024691255.
- Olivier Cappé, Éric Moulines, and Tobias Rydén. *Inference in Hidden Markov Models*. Springer, 2005. doi: 10.1007/0-387-28982-8.
- Jörn Dannemann. Semiparametric hidden Markov models. *Journal of Computational and Graphical Statistics*, 21(3):677–692, 2012. doi: 10.1080/10618600.2012.681264.
- Jörn Dannemann, Hajo Holzmann, and Anna Leister. Semiparametric hidden Markov models: identifiability and estimation. *WIREs Computational Statistics*, 6(6):418–425, 2014. doi: 10.1002/wics.1326.
- Randal Douc and Éric Moulines. Asymptotic properties of the maximum likelihood estimation in misspecified hidden Markov models. *The Annals of Statistics*, 40(5):2697–2732, 2012. doi: 10.1214/12-AOS1047.
- Mathias Drton and Martyn Plummer. A Bayesian information criterion for singular models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 79(2):323–380, 2017. doi: 10.1111/rssb.12187.
- Elisabeth Gassiat and Christine Keribin. The likelihood ratio test for the number of components in a mixture with Markov regime. *ESAIM: Probability and Statistics*, 4:25–52, 2000. URL https://www.numdam.org/item/PS_2000__4__25_0/.

- Elisabeth Gassiat and Judith Rousseau. About the posterior distribution in hidden Markov models with unknown number of states. *Bernoulli*, 20(4):2039–2075, 2014. doi: 10.3150/13-BEJ550.
- Elisabeth Gassiat, Alice Cleynen, and Stéphane Robin. Inference in finite state space nonparametric hidden Markov models and applications. *Statistics and Computing*, 26:61–71, 2016. doi: 10.1007/s11222-014-9523-8.
- Christine Keribin. Consistent estimation of the order of mixture models. *Sankhyā: The Indian Journal of Statistics, Series A*, 62(1):49–66, 2000. URL <https://www.jstor.org/stable/25051289>.
- Roland Langrock, Thomas Kneib, Alexander Sohn, and Stacy L. DeRuiter. Nonparametric inference in hidden Markov models using P-splines. *Biometrics*, 71(2):520–528, 2015. doi: 10.1111/biom.12282.
- Roland Langrock, Timo Adam, Vianey Leos-Barajas, Sina Mews, David L. Miller, and Yannis P. Papastamatiou. Spline-based nonparametric inference in general state-switching models. *Statistica Neerlandica*, 72(3):179–200, 2018. doi: 10.1111/stan.12133.
- Luc Lehéricy. Consistent order estimation for nonparametric hidden Markov models. *Bernoulli*, 25(1):464–498, 2019. doi: 10.3150/17-BEJ993.
- Brian G. Leroux. Maximum-likelihood estimation for hidden Markov models. *Stochastic Processes and their Applications*, 40(1):127–143, 1992. doi: 10.1016/0304-4149(92)90141-C.
- Yong Lin and Mian Huang. Penalized composite likelihood estimation for hidden Markov models with unknown number of states. *Statistics & Probability Letters*, 216:110247, 2025. doi: 10.1016/j.spl.2024.110247.
- Brett T. McClintock, Roland Langrock, Olivier Gimenez, Emmanuelle Cam, David L. Borchers, Richard Glennie, and Toby A. Patterson. Uncovering ecological state dynamics with hidden Markov models. *Ecology Letters*, 23(12):1878–1903, 2020. doi: 10.1111/ele.13610.
- Geoffrey J. McLachlan and David Peel. *Finite Mixture Models*. John Wiley & Sons, 2000. doi: 10.1002/0471721182.
- Théo Michelot, Roland Langrock, and Toby A. Patterson. moveHMM: An R package for the statistical modelling of animal movement data using hidden markov models. *Methods in Ecology and Evolution*, 7(11):1308–1315, 2016. doi: 10.1111/2041-210X.12578.
- Juan Manuel Morales, Daniel T. Haydon, Jacqui Frair, Kent E. Holsinger, and John M. Fryxell. Extracting more out of relocation data: Building movement models as mixtures of random walks. *Ecology*, 85(9):2436–2445, 2004. doi: 10.1890/03-0269.

- Andrew Parton and Paul G. Blackwell. Bayesian inference for multistate “step and turn” animal movement in continuous time. *Journal of Agricultural, Biological and Environmental Statistics*, 22:373–392, 2017. doi: 10.1007/s13253-017-0286-5.
- Jennifer Pohle, Roland Langrock, Floris M. van Beest, and Niels Martin Schmidt. Selecting the number of states in hidden Markov models: pragmatic solutions illustrated using animal movement. *Journal of Agricultural, Biological and Environmental Statistics*, 22(3):270–293, 2017. doi: 10.1007/s13253-017-0283-8.
- Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989. doi: 10.1109/5.18626.
- Judith Rousseau and Kerrie Mengersen. Asymptotic behaviour of the posterior distribution in overfitted mixture models. *Journal of the Royal Statistical Society: Series B*, 73(5):689–710, 2011. doi: 10.1111/j.1467-9868.2011.00781.x.
- Henry Teicher. Identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 34(4):1265–1269, 1963. doi: 10.1214/aoms/1177703862.
- Sumio Watanabe. A widely applicable Bayesian information criterion. *Journal of Machine Learning Research*, 14:867–897, 2013. URL <https://www.jmlr.org/papers/v14/watanabe13a.html>.
- Halbert White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982. doi: 10.2307/1912526.
- Sidney J. Yakowitz and John D. Spragins. On the identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 39(1):209–214, 1968. doi: 10.1214/aoms/1177698520.
- Yudan Zou, Yiqi Lin, and Xinyuan Song. Bayesian heterogeneous hidden Markov models with an unknown number of states. *Journal of Computational and Graphical Statistics*, 33(1):15–24, 2024. doi: 10.1080/10618600.2023.2231055.
- Walter Zucchini, Iain L. MacDonald, and Roland Langrock. *Hidden Markov Models for Time Series: An Introduction Using R*. Chapman and Hall/CRC, 2 edition, 2016. doi: 10.1201/b20790.